

To start defining a pollutant, click the **Add** button and the **Pollutant Definition** window will appear. In the **Pollutant ID** box, you give a unique name for the pollutant (e.g., PM_{2.5}), and then define the characteristics of this pollutant – the **Observation Type** and **Metrics**.

The screenshot shows the 'Pollutant Definition' window. The 'Main' pane on the left has a 'Pollutant ID' field with 'PM2.5', an 'Observation Type' dropdown set to 'Daily', and a 'Metrics' list box containing 'D24HourMean'. Below these are 'Delete', 'Add', and 'Define Seasons for all Pollutant Metrics' buttons. The 'Detail' pane on the right has a 'Metric Name' field with 'D24HourMean', an 'Hourly Metric Generation' dropdown set to 'Fixed Window', and a 'Fixed Window' section with 'Start Hour' (0), 'End Hour' (23), and 'Statistic' (Mean) fields. At the bottom of the 'Detail' pane is a 'Manage Seasons for Individual Pollutant Metrics' section with a list box containing 'QuarterlyMean' and an 'Edit' button. The window has standard Windows window controls at the top right and 'Cancel' and 'OK' buttons at the bottom right.

The **Observation Type** identifies whether a pollutant is measured *Hourly* or *Daily*. In the United States, ozone, sulfur dioxide, carbon monoxide, and others have hourly observations, while particulate matter has daily observations.

Next you need to define a pollutant's **Metrics**. A pollutant has to have one or more metrics, which are daily values calculated directly from daily observations, or through various mathematical manipulations of hourly observations.

To add a **Metric**, click on the **Add** button below the **Metrics** box in the **Pollutant Definition** window. A default name 'Metric 0' will appear in the box. Since the default name is not very descriptive of a metric, it is best to change the name. Typical names used for metrics given in Table 4-4. These are provided just as an example, you may use any names that you like. However, keep in mind that the names that you use for your

metrics need to be consistent with the metric names that you include in your air pollution monitoring and modeling data, as well as your health impact functions. (We will discuss this further below.) Additionally, metric names are used to display pollutant concentrations in BenMAP-CE's mapping window. As such, they must be consistent with GIS naming conventions, meaning they must begin with a letter, and may only contain letters, numbers, and underscores.

Table 4-4. Examples of Metric Names

Name	Description
D1HourMax	Highest hourly value from 12:00 A.M. through 11:59 P.M.
D8HourMax	Highest eight-hour average calculated between 12:00 A.M. and 11:59 P.M.
D24HourMean	Average of hours from 12:00 A.M. through 11:59 P.M.

4.1.2.1 Hourly Metrics

Pollutants that are measured hourly (**Observation Type** = *Hourly*), such as ozone, sulfur dioxide, carbon monoxide, and others, must be characterized by a daily metric, which mathematically summarizes the hourly observations.

Table 4-4 lists some of the ways that metrics can be generated from hourly values. Note that these metrics are not arbitrarily chosen, and instead match the metrics used in epidemiological studies.

The **Detail** section of the **Pollutant Definition** window lets you define the metrics that you want to use. There are three options that you may choose using the **Hourly Metric Generation** drop-down list: *Fixed Window*, *Moving Window*, and *Custom*.

The **Fixed Window** option lets you define simple metrics which are calculated as statistics over a fixed window of hours (**Start Hour** and **End Hour**) within each day. The **Start Hour** should be less than or equal to the **End Hour**, and both can range from 0 to 23, where 0 stands for the period 12:00 am to 12:59 am, and 23 stands for 11:00 pm to 11:59 pm. The **Statistic** includes the *Mean*, *Median*, *Max*, *Min*, and *Sum*. Some examples follow:

- **D24HourMean**: The mean of the observations from 12:00 am through 11:59 pm. **Start Hour** = 0. **End Hour** = 23. **Statistic** = *Mean*.
- **D1HourMax**: The highest hourly value of the observations from 12:00 am through 11:59 pm. **Start Hour** = 0. **End Hour** = 23. **Statistic** = *Max*.
- **D12HourMean**: The mean of the daylight observations, defined as the period from 8:00 am through 7:59 pm. **Start Hour** = 8. **End Hour** = 19. **Statistic** = *Mean*.

Pollutant Definition

Main

Pollutant ID:

Observation Type:

Metrics

- D1HourMax
- D24HourMean**
- D8HourMax
- D8HourMean

Detail

Metric Name:

Hourly Metric Generation:

Fixed Window
 Fixed Window
 Moving Window
 Custom

Fixed Window

Start Hour:

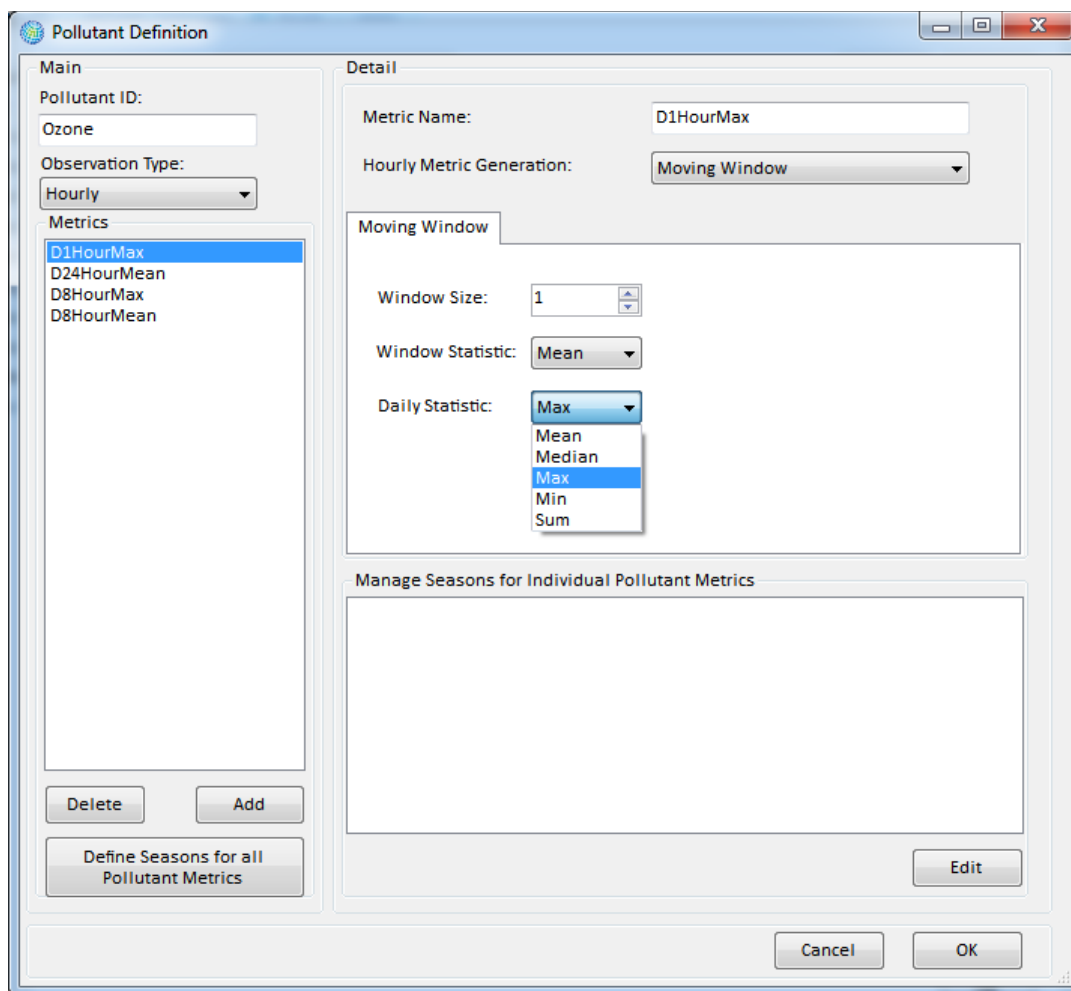
End Hour:

Statistic:

Manage Seasons for Individual Pollutant Metrics

The **Moving Window** option lets you consider metrics that are not based on the same set of hours each day. The **Window Size** defines the number of hours that will be considered together. The **Window Statistic** defines how the hours in the **Window Size** will be characterized. And the **Daily Statistic** defines how BenMAP-CE will use the statistics generated for each window.

For example, consider the highest eight-hour mean (*D8HourMax*) over the course of a day. You would have the following settings: **Window Size** = 8. **Window Statistic** = *Mean*. **Daily Statistic** = *Max*. BenMAP-CE would calculate every possible eight-hour mean, starting with the eight-hour mean from 12:00 am through 7:59 am, and ending with the eight-hour mean from 4:00 pm through 11:59 pm. This would generate 17 possible eight-hour means. BenMAP-CE would then choose the eight-hour mean that has the highest value (see Table 4-2 above for an example).



The **Pollutant Definition** dialog box is used to configure pollutant metrics. It is divided into two main sections: **Main** and **Detail**.

Main Section:

- Pollutant ID:** A text field containing "Ozone".
- Observation Type:** A dropdown menu set to "Hourly".
- Metrics:** A list box containing "D1HourMax", "D24HourMean", "D8HourMax", and "D8HourMean". "D1HourMax" is selected.
- Buttons:** "Delete", "Add", and "Define Seasons for all Pollutant Metrics".

Detail Section:

- Metric Name:** A text field containing "D1HourMax".
- Hourly Metric Generation:** A dropdown menu set to "Moving Window".
- Moving Window Sub-section:**
 - Window Size:** A spinner box set to "1".
 - Window Statistic:** A dropdown menu set to "Mean".
 - Daily Statistic:** A dropdown menu with options: "Max", "Mean", "Median", "Max", "Min", and "Sum". "Max" is selected.
- Manage Seasons for Individual Pollutant Metrics:** A large empty text area.
- Edit Button:** Located at the bottom right of the Manage Seasons section.

Footer: "Cancel" and "OK" buttons.